TEXORA · COURSE NOTES · LECTURE SHEET

Elasticity

Stress, strain and Hooke's law — Young's modulus, bulk and rigidity moduli, Poisson's ratio, the relations between them, and the energy stored in a stretched body.

PHY-1103 · PHYSICS-I · TOPIC 2 · LEVEL 1 TERM 1

1. Deforming force, stress and strain

Solid bodies are not rigid abstractions; under load they change shape or volume, and when the load is removed they may spring back (**elastic** behaviour), partly recover, or keep the new shape (**plastic** behaviour). Elasticity is the physics of the springing back. Two quantities organise it:

- **Stress** — the deforming force per unit area over which it acts: stress = F/A, in pascals (N m^{-2}). For a pull along a wire it is *tensile* stress; for pressure from all sides it is *volume* stress; for forces sliding one face of a block over another it is *shear* stress.
- **Strain** — the resulting fractional deformation: extension/original length for tensile strain, ΔV/V for volume strain, and the angle of shear (in radians) for shear strain. Strain is a pure ratio — no unit.

Within the **elastic limit**, stress and strain are proportional — **Hooke's law** — and the constant of proportionality for each mode of deformation is an **elastic modulus**, a property of the material itself. Beyond the elastic limit the body takes a permanent set, and beyond the breaking stress it fails: the stress–strain curve of a material, with its proportional region, elastic limit, yield point and breaking point, is its mechanical signature — the direct ancestor of the load–extension curves drawn for every yarn and fabric in textile testing.

2. The three moduli and Poisson's ratio

MODULUS	DEFINITION	NOTES
Young's modulus, Y	tensile stress / tensile strain = (F/A) / (ΔL/L)	Governs stretching of wires, rods — and fibres. Steel ≈ 2.0 × 10^{11} Pa.
Bulk modulus, B	volume stress / volume strain = p / (−ΔV/V)	Governs compression under uniform pressure; its reciprocal is compressibility.

Rigidity (shear) modulus, η	shear stress / shear strain = (F/A) / θ	Governs twisting and shape change at constant volume; zero for fluids, which cannot sustain shear.
Poisson's ratio, σ	lateral contraction strain / longitudinal extension strain	A ratio, not a modulus: stretch a rod and it thins; σ ≈ 0.3 for most metals, 0.5 for rubber-like (constant-volume) materials.

WORKED EXAMPLE — STRETCHING A STEEL WIRE

A steel wire 2.0 m long and 1.0 mm² in cross-section (1.0×10^{-6} m²) carries a load of 50 N. Stress = $50 / 1.0 \times 10^{-6} = 5.0 \times 10^{7}$ Pa. With $Y = 2.0 \times 10^{11}$ Pa, strain = stress/Y = 2.5×10^{-4}, so the extension is $\Delta L = 2.5 \times 10^{-4} \times 2.0$ m = 5.0×10^{-4} m = **0.50 mm**. Half a millimetre — measurable with care, which is precisely what the Vernier and Searle's apparatus of the Physics-I sessional are built to do.

3. Relations between the moduli

An isotropic solid has only two independent elastic constants; the four quantities above are tied together by

$$Y = 2\eta\,(1 + \sigma) \quad \text{and} \quad Y = 3B\,(1 - 2\sigma) \quad \text{and} \quad \frac{1}{Y} = \frac{1}{9B} + \frac{1}{3\eta}$$

These relations are also a consistency police: from the second, σ cannot exceed 0.5 (else Y would go negative), and $\sigma \to 0.5$ means $B \gg Y$ — a material that stretches easily but refuses to change volume, which is rubber exactly. Given any two constants, the rest follow; examination questions are usually just this substitution. A torsion application worth meeting here: a wire twisted by a couple stores its stiffness in η, and the torsional pendulum of the sessional course measures η dynamically from the period, $T = 2\pi\sqrt{(I/c)}$ with $c = \pi\eta r^4/2L$ the torsional constant of the wire.

4. Energy stored in a stretched body

Stretching a body does work against its internal restoring forces, and within the elastic limit the work is banked as **elastic potential energy**, returned on unloading. Since the force grows linearly from 0 to F (average F/2) through an extension x,

$$U = \tfrac{1}{2} F x = \tfrac{1}{2} \times \text{stress} \times \text{strain} \times \text{volume}, \quad \text{i.e. energy density } u = \tfrac{1}{2} \times \text{stress} \times \text{strain} = \sigma^2/(2Y)$$

WORKED EXAMPLE — ENERGY IN THE WIRE OF §2

$u = \frac{1}{2} \times (5.0 \times 10^{7}) \times (2.5 \times 10^{-4}) = 6.25 \times 10^{3}$ J m^{-3}. The wire's volume is $2.0 \times 1.0 \times 10^{-6} = 2.0 \times 10^{-6}$ m^3, so $U = 6.25 \times 10^{3} \times 2.0 \times 10^{-6} =$ **0.0125 J** — equivalently $\frac{1}{2} \times 50$ N $\times 5.0 \times 10^{-4}$ m. Both routes agree, as they must; the two forms of the formula are one fact in two clothes.

The same energy density, pushed to the elastic limit, gives the **modulus of resilience** — the energy a material can absorb and return, the property that makes spring steel springy and that decides how much abuse a fibre can take in processing and spring back from.

5. The textile thread

Textile testing is elasticity with trade names. A fibre's **initial modulus** is Young's modulus measured on its stress–strain curve; its **elongation at break** is the failure strain; its **elastic recovery** — how much of an imposed strain returns on unloading — is what keeps a garment's shape, and the near-perfect recovery of elastane (σ close to the rubber limit, enormous elastic range) is why a few percent of it transforms a fabric. Yarn elasticity governs tension peaks on the loom and the knitting machine: a yarn that can store and return the energy of §4 survives the shed and the needle; one that cannot, breaks. The moduli also silently set fabric hand, drape and crease recovery — this topic's physics, judged by every customer's fingers.

6. Glossary — key terms of this topic

TERM	MEANING
Stress	Deforming force per unit area (Pa). Tensile, volume or shear according to the loading.
Strain	Fractional deformation (ΔL/L, ΔV/V, or shear angle); dimensionless.
Hooke's law	Within the elastic limit, stress is proportional to strain; the constant is the relevant elastic modulus.
Young's modulus (Y)	Tensile stress ÷ tensile strain; the stiffness of a material in stretching.
Bulk modulus (B)	Uniform pressure ÷ (fractional volume decrease); inverse of compressibility.
Rigidity modulus (η)	Shear stress ÷ shear strain; resistance to change of shape.
Poisson's ratio (σ)	Lateral contraction strain ÷ longitudinal extension strain; between 0 and 0.5 for real materials.
Elastic limit	The greatest stress from which a body fully recovers on unloading.

7. Key points and summary

- Stress (F/A) and strain (fractional deformation) give Hooke's law up to the elastic limit; each deformation mode has its modulus — Y, B, η — and Poisson's ratio links extension to lateral contraction.
- The moduli are interlocked: $Y = 2\eta(1 + \sigma) = 3B(1 - 2\sigma)$, with $1/Y = 1/(9B) + 1/(3\eta)$; $\sigma \leq 0.5$, rubber sitting at that limit.
- Stored elastic energy density is ½ × stress × strain — the basis of resilience, springs and fibre survival in processing.

8. Practice

Answer in your own words or with full working; brief model answers follow.

1. A wire of length 1.5 m and area 2.0 mm² extends by 0.90 mm under a load of 120 N. Find Young's modulus of the material.
2. A material has $Y = 1.2 \times 10^{11}$ Pa and $\sigma = 0.25$. Find its rigidity modulus η.
3. The same material (question 2) — find its bulk modulus B from Y and σ.
4. A steel rod is stretched and its diameter measurably thins. Name the ratio that quantifies this and state its theoretical maximum value, with the material class that approaches it.
5. Compute the elastic energy stored per unit volume in a sample under a stress of 4.0×10^{7} Pa in a material of $Y = 1.0 \times 10^{11}$ Pa.
6. A fluid is said to have rigidity modulus zero. Explain why, and what this means for its shape.
7. Two yarns have the same breaking load; yarn A has twice the initial modulus of yarn B. Under a small working tension, which yarn extends more, and why does that matter on a knitting machine?
8. State Hooke's law and name the point on the stress–strain curve beyond which a wire (a) no longer recovers fully, (b) breaks.

Answer key

1. Stress = $120 / 2.0 \times 10^{-6} = 6.0 \times 10^{7}$ Pa; strain = $0.90 \times 10^{-3} / 1.5 = 6.0 \times 10^{-4}$; $Y = 6.0 \times 10^{7} / 6.0 \times 10^{-4} = 1.0 \times 10^{11}$ Pa.
2. $\eta = Y / [2(1 + \sigma)] = 1.2 \times 10^{11} / 2.5 = 4.8 \times 10^{10}$ Pa.
3. $B = Y / [3(1 - 2\sigma)] = 1.2 \times 10^{11} / [3 \times 0.5] = 8.0 \times 10^{10}$ Pa.
4. Poisson's ratio; maximum 0.5, approached by rubber-like, volume-preserving materials.
5. $u = \sigma^2/(2Y) = (4.0 \times 10^{7})^2 / (2 \times 10^{11}) = 1.6 \times 10^{15} / 2 \times 10^{11} = 8.0 \times 10^{3}$ J m^{-3}.

6. Rigidity resists change of shape under shear; a fluid cannot sustain a shear stress at all — it flows instead — so $\eta = 0$ and it has no shape of its own, taking the shape of its container.
7. Yarn B (half the modulus) extends twice as much at the same stress, since strain = stress/Y. On the machine, the stretchier yarn gives larger tension swings and loop-length variation — a levelness and breakage concern.
8. Hooke's law: within the elastic limit, stress $\propto$ strain. (a) The elastic limit — beyond it a permanent set remains. (b) The breaking stress (ultimate tensile stress region) — beyond it the wire fractures.
